

CSC 4132 – Project

ECC Implementation

Deadline: Should present by first week of March

Project Idea: “Cryptographically Secure Government e-Procurement System (CSePS)”

A mini system that uses **ECC cryptography, digital signatures, timestamping, and hashing** to guarantee **bid integrity, bidder anonymity (until deadline), non-repudiation, and verifiable fairness** in a procurement process.

Project Goal

Build a prototype procurement workflow where:

- Bidders **submit bids encrypted** so evaluators cannot read them early.
- Every bid is **digitally signed** so authenticity and non-repudiation are guaranteed.
- A **public, tamper-proof ledger/log** stores all submissions (hash-based), ensuring transparency.
- Bids can only be **decrypted after the deadline** using a shared threshold key or multi-party decryption.
- Everyone can verify that bids were not modified or opened prematurely.
- This solves long-standing real-world issues: early bid leaks, bid tampering, corruption, and denial disputes.

Group information

Group 01	S/20/421	S/20/535
Group 02	S/20/534	S/20/381
Group 03	S/20/504	S/20/302
Group 04	S/20/515	S/20/380
Group 05	S/20/139	S/20/109
Group 06	S/20/140	S/20/422
Group 07	S/20/196	S/20/545
Group 08	S/20/369	S/20/594
Group 09	S/20/335	S/20/314
Group 10	S/20/570	S/20/455